The Published P-256 Seed Was Not the First Viable Candidate: A Preregistered Verification for P-256

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***Abstract*—Anthropic reported viable candidate 876 before P-256's seed, candidate 1,842. Within their reconstructed stream we verify the skip: our preregistered confirmation enumerating ten runs' prefixes and testing the mundane rules that could explain it—none tested does; two arms untested. No weakness in P-256 is shown; no action is needed.**

***Index Terms*—elliptic-curve cryptography, NIST curves, P-256, verifiable randomness, cryptographic standards.**

In brief. The P-256 elliptic curve protects much of the world's encrypted traffic. Within Anthropic's reconstructed stream, its seed—the starting number the curve is derived from—was the 1,842nd candidate produced; we verified that candidate 876 passes every check we recomputed, yet was not published—and no tested criterion or rule explains the choice. Viability is tested under two author-defined screens: one assembled from the documented conditions, and a stricter superset tier of our construction. This shows no weakness in P-256; readers need take no action. What we take on trust: Anthropic's reconstruction of the timestamp and generator stream—if wrong, the ordering claims fall, but the seed-to-curve derivation stands. Untested: per-seed invocations, a wall-clock race, crash-and-restart, and Valsorda's second criterion. Terms: a *candidate* is one seed in the sequence; *viable* means it passes the generation checks; *preregistered* means our checks were frozen before our own P-256 computation; candidate 876's skip was already public, and the premises and one voided P-224 pass predate the freeze.

INTRODUCTION

For over two decades, the seeds behind the NIST elliptic curves arrived with no explanation of where they came from. These curves—P-192, P-224, P-256, P-384, P-521—underpin an enormous share of the world's encrypted traffic. The vacuum attracted speculation for years, most colorfully in Jerry Solinas's 2015 recollection—preserved only in secondary accounts—of a hashed passphrase along the lines of "give Bob and Jerry a raise" [3]. Filippo Valsorda organized a bounty for the mystery, US$36,864, donated to the Internet Security Research Group [10], [3].

In October 2026, researchers at Anthropic claimed it: an internal AI model traced all five NIST prime-curve seeds in under half an hour to the `lrand48` generator shipped with 1990s Unix C libraries, each invocation seeded with a Unix timestamp [3]. The finding was reassuring in one sense—the seeds came from a mundane, reproducible process—but it opened a sharper question: knowing *how* the candidates were produced does not tell us *which candidate* was chosen, or why.

Anthropic's reconstruction showed that the published seeds were not always the first viable candidates of their runs: the published P-256 seed is candidate 1,842 of its run, while candidate 876 was also viable [3]. If the generation process took the first candidate that yielded a suitable curve, 1,842 would not have been published: under the loose viability tier the earliest viable candidate is 78, and under the strict tier it is 876. Neither was published. The headline rests on 876 because it is the strongest case: it passes the strict tier, while 78 qualifies only at the loose tier, on a cofactor of 4. Both figures come from the completed enumeration of all ten runs' prefixes (Result IV): exactly eight candidates are viable under the loose tier before 1,842. Something, whether mundane or deliberate, selected among them—or the period process never evaluated them in stream order at all—and nobody knows which. One candidate-rejection rule—a bit-length rejection reported independently by Valsorda [10]—has been identified; the rest of the selection process remains unresolved.

Why does a skip matter? Verifiable randomness means the seed commits to the curve with no room for choice. A documented selection rule restores that; an undocumented skip reopens exactly the room for hidden selection the design was meant to close. The demonstrated room in this instance: within the reconstructed prefix, the P-256 choice was among two strict-tier candidates (one bit) or nine loose-tier ones, the published candidate included (about 3.2 bits)—a count of prior viable candidates in the prefix, not a bound on any wider search's freedom. That is what we tested. The first-viable search we test is our reconstruction of the natural selection rule, not a rule NIST or X9.62 documented. And "verifiable randomness" is used in the standards' narrower sense—anyone can re-derive *the* published curve from *its* published seed, which holds here—not as a guarantee about how the seed itself was chosen.

We report an independent, preregistered confirmation audit of that selection. We reconstructed the seed-to-curve pipeline from the published standards and verified its seed-to-b derivation byte-for-byte against every published prime-curve selection; the coefficient a is fixed at $p-3$ by the standard. Our headline result is a verified refutation: the published seed was *not* the first viable candidate. Candidate 876 has prime order, passes even the strictest criteria we apply, and precedes the published candidate—yet was passed over.

We are careful about what this does and does not establish; the three hedges that bind every claim are stated once, in Scope and Limits. The preregistered programme is now run: we report exactly what we verified, mark the arms that remain unevaluated, and name the selection rule what the completed tests support—not explained by the documented criteria tested so far. The work also proposes a template for future audits, exercised here in its frozen tiers, pre-committed negatives, complete enumeration, and rule battery.

Background: Seeds Without a Story

Start with what a seed does. In the generation method later codified as FIPS 186-2 (published January 2000; the curves were generated at the NSA in 1997–98 for the draft X9.62 text it standardized) [14], [1], the seed is the input to a deterministic recipe, in four steps. (1) Hash-expand the seed with SHA-1 (appendix A.6.4) to obtain c. (2) Set $b^2 = -27/c \pmod p$ and take a square root to get b—the step at which this audit's shortcut implementation later broke on P-224. (3) Check the verification equation $b^2c \equiv -27 \pmod p$ (A.6.5); failures are discarded—and the check cannot distinguish b from $-b$. (4) Accept the curve only if its order—the number of points on it—has the required form: in the general X9.62 form, a large prime factor with small cofactor; for these five NIST curves, the strongest form—the order itself prime (cofactor $h = 1$). The seed is the recipe's only arbitrary input; everything else follows mathematically. But the design assumes the seed itself is above suspicion—and for 28 years, nobody could say where the seeds came from.

Anthropic's October 2026 report closed the origin question with an anticlimax: no hashes, no passphrases. Each seed is 20 bytes drawn as the top eight bits of twenty successive `lrand48()` outputs, from a generator seeded once with a Unix timestamp; candidates are numbered sequentially, twenty bytes at a time [3]. Exposing the top of every state is the fingerprint: the retained bytes, tied together by the generator's known linear recurrence, let a lattice-reduction (LLL) computation—the classical method for truncated congruential generators—recover the full 48-bit states, and every recovered state regenerates all 160 bits of its seed. The five NIST prime curves map to timestamps between May 1997 and April 1998, and five ANSI X9.62 example curves match the same pattern [2], [3]. The base points of P-192 and the X9.62 curves were traced separately to a generator inside FreeLIP [3].

But origin is not selection. Anthropic's inventory records skipped viable candidates: P-256's published seed is candidate 1,842 while candidate 876 was viable; the P-224 run passed over three viable candidates, and the X9.62 runs seven more [3]. Under this audit's preregistered tiers the same runs show more skips (Result IV)—the counts differ because the viability criteria differ. Independently, Valsorda reproduced the generation process and reported two additional apparent criteria beyond X9.62—one an inferred bit-length rejection rule, one still unexplained [10]. This is the gap our audit addresses: given the verified generation process, was the published seed simply the first viable candidate? The question is falsifiable—recompute the orders—and it is the exact place a hidden selection would leave a trace.

The question matters beyond historical curiosity. Trust in shared parameters rests on verifiable randomness—the idea, championed by the SafeCurves project [8]—that parameters should be generated by a documented, reproducible process with no room for hidden selection. When the NIST curves' provenance was opaque, the community split: some accepted the parameters on authority, while others—notably the Brainpool designers, whose generation procedure is published [12]—insisted on documented generation to eliminate the selection step this article audits. The Dual EC episode showed what opacity costs at its worst: Shumow and Ferguson showed in 2007 its constants could conceal a backdoor known only to whoever chose them [13], and its later practical exploitation in TLS implementations (Checkoway et al.) [9] retroactively poisoned trust in the process. That selection among acceptable candidates is a real attack surface was shown by the BADA55 team: they generated over a million curves, each with a "verifiably random" Brainpool-style explanation, to show such explanations can cloak curves chosen for hidden properties—and called the "verifiably random" terminology itself deceptive [15]. The defence of the NIST process is Koblitz and Menezes's: a seed search for a hidden weakness was beyond 1997 capacity, they argued, and a weak class would have surfaced in decades of cryptanalysis—yet they record Bernstein and Lange's counterpoint: tens of bits of selection freedom could still target a rare weak class [14]. Our finding sits between these poles: a verified mundane process with one unexplained selection. It is neither an exoneration nor an accusation, and the article is written to keep it that way. New standards will keep asking users to trust generated parameters, and the community needs templates for auditing verifiable-randomness claims, including ones that report honest partial results.

Method: A Preregistered Forensic Audit

Forensic claims are easy to make and hard to check, which is why we preregistered the audit before executing it. Hypotheses, search space, viability definitions, controls, stop rules, and a compute cap were frozen in a preregistration document [11] that survived two rounds of independent methods review. Without a frozen document, an audit can always be re-tuned, consciously or not, until the data confesses. Every deviation below was disclosed rather than folded in silently, and every negative outcome had a pre-committed interpretation, including the outcome "selection rule not explained by the tested criteria."

The preregistration was frozen three times on October 17, 2026: the initial freeze, a second freeze incorporating 18 adjustment items from the first round of independent methods review, and the amended re-freeze cited above, incorporating a further 19 items from the second round. The chronology, stated exactly: the premises the audit builds on—Anthropic's seed regeneration, re-implemented locally—were established before the freezes and are listed as premises in the deposited document [11]; the freezes predate every computation of candidate orders, viability values, and counts reported here. The voided pass's P-224 counts—identical to the rerun's: the shortcut's value is always a true root or its negation (its square is the radicand times its Legendre symbol), and the two roots' curves are isomorphic (below)—were the only candidate orders computed before the first freeze; no P-256 quantity was. Pipeline construction, including the square-root correction below, proceeded against the published selections (the ground-truth anchors); its one voided pass—P-224's first enumeration under the defective shortcut—was discarded and rerun under the corrected method, both logged as deviations in

the deposited reconciliation. The chronology matters, and we state it plainly: Anthropic and Valsorda had already published candidate 876's skip before the freeze, so for that candidate this audit is a preregistered confirmation—hypotheses, tiers, and interpretations fixed before we computed anything—not a discovery. The freeze binds our computation and interpretation; the question itself was already public.

Pipeline reconstruction. For each candidate n in a run, we regenerate the 20-byte seed from the `lrand48` stream (`srand48`(t) once per run; twenty `lrand48()`$\gg$23 draws per candidate), derive b via FIPS 186-2 A.6.4, and verify via A.6.5. One correction emerged: a square-root shortcut valid only for primes $p \equiv 3 \pmod 4$ fails on P-224's prime ($1 \bmod 4$); we replaced it with the general Tonelli–Shanks method [5]. The derivation cannot distinguish b from $-b$—both satisfy A.6.5, and nothing in the standards fixes which representative a generator outputs. The sign is fixed once per run by the published coefficient—the anchor—not fitted per candidate: under the other sign, the run's own published candidate fails to reproduce the published curve. For the nine runs whose prime is $3 \pmod 4$, the root is the exponentiation root $(-27/c)^{(p+1)/4}$, a deterministic function of the seed alone. A.6.4 fixes only the verification equation, not the representative, so the root's identity is an empirical claim; its evidence is the anchor—every such run's published coefficient matches it—and a generator using the other root lies outside the process this audit tests. Only P-224's prime is $1 \pmod 4$, where a root routine could return either representative; the choice cannot affect any count, because for $p \equiv 1 \pmod 4$ the two roots' curves are isomorphic over $\mathbb{F}_p$—$(x, y) \mapsto (-x, sy)$, $s^2 = -1$ existing exactly then—hence have the same order for every candidate. The anchor's negated-root convention is needed only to reproduce the published coefficient byte-exactly. For the P-256 run, the stream timestamp is $t = 871{,}696{,}354$ (1997-08-16 01:52:34 UTC, Anthropic's reconstruction). Curve orders in the enumeration were computed with PARI/GP's `ellcard` [4]. The second algorithm, `ellsea` (Schoof–Elkies–Atkin), was not run on all 31,625 candidates; it has since been run, under PARI/GP 2.15.4, over every viable candidate's order in all ten runs—199 orders, zero disagreements—and over the P-256 run's whole prefix; the remaining bulk orders are `ellcard`'s alone, spot-checked as Result IV states—and the headline orders were verified outside PARI/GP altogether (Result I).

Viability tiers. The order is the number of points on the curve; the cofactor h measures how far that count is from being prime ($h = 1$ means the order itself is prime—the form all five NIST curves take); and the embedding degree guards against a class of attacks that reduce the discrete-log problem to a weaker setting. A candidate is viable under `V_loose` if its order $N = h \cdot q$ with q prime, $q \neq p$ (excluding anomalous curves, whose order equals the field prime), and cofactor $h \leq 8$ (an arbitrary loose bound). The strict tier `V_strict` pins $h = 1$ and requires embedding degree $k > 20$ (the MOV condition [6], applied as the test $p^d \not\equiv 1 \pmod q$ for $1 \leq d \leq 20$)—our conservative superset of the standards' requirements, stricter than anything the 1997 process is documented to have applied. Its predicates map to Background's documented conditions—seed verification, non-anomalous order, a large prime factor with small cofactor, the MOV bound—plus prime order itself, the property all five published curves in fact take, frozen as our retrospective tightening before any order was computed. `V_loose` does not test the embedding degree, so a loose-tier "viable" is a pass of a cofactor-and-primality screen; every candidate the headline turns on is additionally checked at the strict tier. Primality in the pipeline is decided by PARI/GP's `isprime`, applied to the prime factor q of every candidate's order: it returns proven primes—a deterministic test, not a probabilistic screen—and the "proof-grade" label in Table I refers to it; the checkable artifact the proof itself does not leave is `primecert`'s, below. As a separate cross-check, the q of every skipped-viable candidate (189 across the ten runs) was re-tested under five pinned Miller–Rabin variants [7] —base sets $\{2\}$, $\{2, 3\}$, and $\{2, 3, 5, 7, 11\}$, the last in three forms: plain, and preceded by trial division to 10^3 and to 10^6—plus `isprime` again: all 189 test prime under every variant (Result IV). Extracting q and h is mechanical: trial division by all primes up to 10^6 accumulates h (stopping once $h > 8$); a prime remainder is q, a composite remainder means no single prime factor at that cofactor and the candidate is non-viable. The embedding degree is the multiplicative order of p modulo q, tested for every $h = 1$ candidate by checking $p^d \not\equiv 1 \pmod q$ for all $1 \leq d \leq 20$. We also generated PARI/GP `primecert` certificates for the prime factor of every viable candidate—139 distinct values covering all 199 viable rows (the X9.62 example runs share stream prefixes over the same fields, so the same candidate can recur across runs with the same q: 96 values cover one row each, 26 cover two rows, 17 cover three—60 repeat rows, none within a single run)—verified each, and archived them with the reproduction package. Several verification elements here—these certificates, the Result IV spot check, and the Controls—were added in revision, after the freezes: post-registration corroboration, reported as such, changing no preregistered outcome. The choice $h \leq 8$ does not drive the P-256 result: the eight skipped candidates carry cofactors 4, 3, 1, 7, 6, 2, 8, 5 in run order, so under every bound from 1 to 8 some viable candidate still precedes 1,842 (876 at bounds 1–2, 186 at 3, 78 at 4 and above). Non-singularity needs no separate test: a singular cubic yields no order—the computation errors and is logged, not counted; none occurred, and the only order errors logged among the 31,625 were P-521's 51 timeouts, all retried to completion.

The tiers matter because "viable" is doing load-bearing work: a skip is only interesting if the skipped candidate genuinely qualified.

Controls. Three controls anchor the machinery. First, a composite-order control curve, built deliberately over P-192's field with $b = 1$, is correctly rejected at the viability stage: its order $N = 627710173538668076383578942315861295102391618412398762800 1$ is composite (both order algorithms agree), and the pipeline rejects it on the composite-remainder branch ($h = 1$, remainder composite)—the viability test is not a rubber stamp. Second, the orders of published curves recompute byte-identically

(P-256's matches FIPS 186-2), anchoring our arithmetic to ground truth. Third, every PRNG implementation was validated against published reference vectors, and the binary-curve search machinery was calibrated by recovering known seeds (Result III). One gap in the original design, stated plainly, was closed in revision: the anomalous, cofactor, and embedding-degree branches had no dedicated synthetic control. Known-answer controls were therefore run through the production predicate code over small prime fields (archived with the reproduction package): an anomalous curve, its order equal to its field prime, is rejected on the anomalous branch; a curve of order $8q$ passes at the cofactor boundary, one of order $648q$ is rejected, and one of order $9q$—the first rejected cofactor—is rejected; and the embedding-degree test fails prime-order curves of known embedding order 10 and exactly 20, the strict tier's boundary, and passes one of order 21 and one of order 3,078. All eight logic controls behave as required. (The rule battery's planted-rule control was also run in revision: six rules planted on three runs' real viable sets—18 plantings, all detected—and none on three rule-free synthetic sets—0 of 3, a false-positive bound of ≈63%—uninformative.) Compute was capped as the preregistration specifies (Scope and Limits). As a cross-version check on the October 2026 completion run (PARI/GP 2.19.0), the load-bearing orders were recomputed under the audit's validated PARI/GP 2.15.4—every viable P-224 candidate (58), all 51 retried P-521 candidates, and P-521's two viable candidates, 242 and 1,136 ($58+51+2=111$; the two viable are disjoint from the 51, which all resolved non-viable)—with `ellcard` and `ellsea` agreeing in all 111 cases. What produced the published seed acted after the last stage of this pipeline: a selection among the viable candidates, or a process that never enumerated them—no record states which.

Result I: The P-256 Refutation

The first-viable hypothesis fails for P-256, verified end to end downstream of the reconstructed stream. (Results II–III establish the pipeline and the machinery controls behind that verification.) Candidate 876 of the P-256 run has the order, cofactor, and primality properties shown in Table I[1]: order N prime (proof-grade; see Method), cofactor $h = 1$, $q = N \neq p$ (non-anomalous), with N, p, and q all 256 bits. Candidate 876 passed the A.6.5 seed-to-curve verification ($b^2c \equiv -27$); its membership in the run is established by the stream regeneration described in Method, not by this check alone. Its order was computed by two distinct algorithms within PARI/GP (`ellcard` and `ellsea`) in agreement; the independent anchor is the exact match of the published candidate's recomputed order to the documented NIST value. Outside PARI/GP entirely, the orders of 876, 78, and 1,842 were also verified by direct point arithmetic in pure Python: at the first three x-coordinates yielding points on the curve (each checked non-identity and on-curve), $N \cdot P = \mathcal{O}$ with N prime forces $\#E = N$: a non-identity point annihilated by the prime N has order exactly N, so N divides $\#E$, and the Hasse interval (which contains N) admits no other multiple of N; for 78, one tested point satisfies $(2q)\cdot P \neq \mathcal{O}$ and $4\cdot P \neq \mathcal{O}$ in addition to $(4q)\cdot P = \mathcal{O}$, leaving $4q$ as its only possible order ($4q$ divides $\#E$; $8q$ lies outside the interval), and $\#E = 4q$. The check shares its inputs (seed, and b derived from it) with the pipeline, so it rules out order-computation error, not a common-mode seed or stream error—which the byte-exact published-candidate anchor covers instead. The published candidate, $n = 1{,}842$, has order exactly equal, as an integer, to the documented NIST value in FIPS 186-2 [1]. Since $876 < 1{,}842$ and both are curve-viable under the preregistered tiers—preceding assumes the validated `lrand48` stream model extends unchanged to candidate 876—the published seed was not the first viable candidate—within Anthropic's reconstructed timestamp and indexing, the premise Scope and Limits states; the seed-to-curve derivation itself is verified unconditionally. Candidate 78, the earliest under `V_loose`, is documented the same way[2]. The refutation is deterministic: two order computations suffice, both replayable from the archived materials. Nor does the headline depend on the uniform choice of root representative: under the negated root, all three twists are non-viable (each twist order factors with no cofactor ≤ 8 leaving a prime quotient; full factorizations archived). No standard was violated by the skip—the first-viable rule is our reconstruction of the natural selection rule, and what fails is that reconstruction, not any procedure NIST or X9.62 documented. Both ran under PARI/GP 2.15.4 [4]; the `ellsea` computations cost 20–30 seconds per 256-bit order. (Anthropic's inventory reported this skip; our contribution is its independent, preregistered verification.)

TABLE I
WHY CANDIDATE 876 COUNTS AS CURVE-VIABLE UNDER THE AUDIT'S TIERS: ON EACH CHECKED PREDICATE BELOW, THE SKIPPED CANDIDATE MATCHES THE PUBLISHED ONE. EMBEDDING DEGREE IS REPORTED AS THE TESTED BOUND: $p^d \not\equiv 1 \pmod{q}$ FOR ALL $d \leq 20$ (METHOD).

Property	$n = 876$ (skipped)	$n = 1{,}842$ (published)
Order is prime	Yes (proof-grade; see Method)	Yes (proof-grade; see Method)
Cofactor h	1	1
Non-anomalous ($q \neq p$)	Yes	Yes
Embedding degree $k > 20$	> 20	> 20
A.6.5 seed verification	Pass	Pass
Valsorda's reported bit-length criterion (inferred rejection rule)	Not rejected	Not rejected

The cheapest innocent explanation does not rescue the hypothesis. Candidate 876 passes even `V_strict` (Table I). It also passes Valsorda's reported bit-length rejection rule—a criterion inferred in his reproduction, not a documented standard requirement: reject any candidate whose order is longer than the field prime, $\mathrm{bitlen}(N) > \mathrm{bitlen}(p)$—since $\mathrm{bitlen}(N_{876}) = 256 = \mathrm{bitlen}(p)$, so it is not rejected: the

[1]Candidate 876's seed is `50C71A63F911DB7B2F31618ECF609D37154437F9`; its order is N_{876} = 1157920892103562487626974469494075735298 98027617956640771784107778367342146894 3.

[2]Candidate 78's seed is `DE9F9A343DFA8F3B009D335319C20276AD62EEA5`; its order N_{78} = 1157920892103562487626974469494075735294886541462554594848136478261825732337 08 = $4q$ with q prime, checked with `isprime`—its cofactor $h = 4$ excludes it from `V_strict`.

skip is not explained by that filter either. Whatever excluded candidate 876, it was not the order, cofactor, or embedding-degree requirements as stated in the standards, nor that one independently reported extra filter. (A period primality test could still have failed 876 only by being buggy or nonstandard, or by abandoning its proof unfinished: 876's order is itself prime and certified, and no completed primality test rejects a true prime—the certificates (Method, Result IV) close the first two routes; the third is unevaluated, with the timing arms of Result IV.) Under our preregistered outcome table, the audit then proceeded to the battery of mundane selection rules reported in Result IV; since none reproduces the published selection, the pre-committed outcome applies: "selection rule not explained by the documented criteria tested so far"—a checkable negative, not a failure.

What could the rule have been? Several mundane mechanisms fit the evidence so far, and the audit was designed to distinguish them rather than to presume. The implementer (our label for whoever ran the 1997 generation—an assumption about the process, not a finding) may not have tested candidates in generation order—a period program might have scanned backward, or strided, or raced parallel workers and taken the first finisher. The period viability test may have been weaker than ours, or too slow on period hardware for some candidates. Valsorda's second, still-unexplained criterion [10] remains live throughout: neither source states it in checkable form—what quantity it constrains goes unreported—so the battery cannot test it, nothing here excludes it, and whether 876 or 78 satisfy it cannot be evaluated. The rule may have been batch-then-pick, or—the possibility the preregistration refuses to exclude—a human may simply have picked. "Not explained" is not a euphemism for "suspicious": most mundane rules, if confirmed, would close the question as decisively as a backdoor finding would open it. The battery in Result IV tests the hypotheses the stored data can decide and names the rest unevaluated.

Result II: The Pipeline Verifies

Before the refutation means anything, the reconstruction itself must be trustworthy. For all ten published prime-curve selections—the five NIST curves and five X9.62 example curves—the seed-to-b derivation reproduces the published coefficient byte-exactly: nine under this implementation's Tonelli–Shanks root, and P-224 under the negated root (both roots satisfy the A.6.5 verification equation, so this is a choice of representative). Each published candidate index (P-256: 1,842; Table II lists all ten) regenerates its published seed and coefficient through the single pipeline: ten reproductions, zero mismatches.

Result III: A Control—The Machinery Finds What It Should

Before trusting the refutation, we checked the machinery on the four binary-curve seeds whose provenance is already explained. For the binary curves, the generator was seeded not with a full Unix timestamp but with the timestamp divided by two (a "halved clock," a detail of the binary-curve tooling). We recovered each seed by exact stream inversion: the 2^{17} pre-image candidates of the first 31-bit output are walked back through the LCG, and a state counts only if forward regeneration reproduces the seed in full. All four explained binary seeds (B-233, B-283, B-409, B-571) were recovered. The recovered generator states (at candidate positions 155, 32, 60, and 73) span 15:24 to 15:32 UTC on 17 April 1998. Each curve came from its own invocation, minutes apart. This section is a control on the machinery, not a result about the prime curves.

Recovering all four validates the stream-inversion machinery (generator recovery, not the prime-curve order counting, anchored separately in Method): a null result from it will be interpretable as evidence about the world, not a methods artifact. The multiple-comparisons exposure of the exact-match step is negligible: each search walked back 2^{17} pre-image candidates per seed, and a recovery requires regenerating all 160 bits of a seed, so the expected number of chance hits across the four inversions is bounded by $4 \times 2^{17} \times 2^{-160}$—effectively zero.

Result IV: All Ten Runs, and the Mundane-Rule Battery

The enumeration is complete: 31,625 candidates across the ten runs, each from its first candidate through its published one. P-224's enumeration, voided in its first pass by the square-root bug of Method, was re-run: all 7,026 candidates resolved; the published seed is the run's last candidate and its 58th `V_loose`-viable one, with 57 passed over and 11 under `V_strict`. P-521's 51 candidates that timed out at the original 300-second cap were retried at a 30-minute timeout: all resolved, every one non-viable, confirming candidate 242 (cofactor 6) as P-521's first viable and only loose-tier skip; since 242 fails `V_strict`, the run's first strict-tier candidate is its published one. The enumeration's remaining orders are `ellcard`'s alone; one further check was added in revision—a stratified sample of 32 archived non-viable candidates (four per run over the eight original-pipeline runs, drawn uniformly at random within each run; the other two runs are the October completion runs) recomputed under `ellsea` agreed with the archived `ellcard` order in all 32 cases. Zero disagreements in 32 draws bounds the disagreement rate only coarsely (95% upper bound ≈9%), and we report it as that, not as verification of the bulk: the per-run counts of Table II for the other eight sampled runs are single-algorithm results subject to that bound; the completion runs' non-viable bulk is likewise single-algorithm and outside the sample, while every viable candidate's order is dual-verified (Method) and its prime factor carries a certificate.

Over these results we ran the mundane-rule battery—twelve pre-committed rules, (a rule explains the data only if it reproduces the published selection in every run); Table III enumerates all twelve. The twelve are the selection rules; the primality and transcription arms and the strict-tier filter reported below are separate preregistered arms, not among the twelve. The rules are defined to the bit: candidates are numbered from 1; a strided scan's phase ϕ selects viable candidates with $(n-1) \bmod s = \phi$ and picks the first; lexicographic rules

TABLE II
THE COMPLETED ENUMERATION, PER RUN: CANDIDATES EXAMINED, THE PUBLISHED CANDIDATE'S INDEX, THEN UNDER EACH TIER THE FIRST VIABLE CANDIDATE AND HOW MANY VIABLE CANDIDATES PRECEDE THE PUBLISHED ONE ("SKIPS"). EVERY RUN'S PUBLISHED CANDIDATE IS ITS LAST BY CONSTRUCTION. THE `V_LOOSE` COLUMNS REPORT METHOD'S COFACTOR-AND-PRIMALITY SCREEN (EMBEDDING DEGREE UNTESTED AT THAT TIER) AND INHERIT THE $h \leq 8$ BOUND (SENSITIVITY: METHOD).

Run	Candidates	Published	`V_loose`		`V_strict`	
			First	Skips	First	Skips
P-192	139	139	14	3	139	0
P-224	7,026	7,026	77	57	77	11
P-256	1,842	1,842	78	8	876	1
P-384	2,688	2,688	325	9	1,497	2
P-521	1,136	1,136	242	1	1,136	0
prime192v2	6,013	6,013	14	38	139	11
prime192v3	5,228	5,228	14	28	139	5
prime239v1	2,185	2,185	6	12	36	2
prime239v2	2,320	2,320	6	13	36	3
prime239v3	3,048	3,048	6	20	36	7

compare the 20-byte seeds in byte order; Hamming weight counts the seed's 1-bits, ties broken to the smaller candidate number; the median is the lower middle element of the viable list in run order. The ten runs are not independent streams—the X9.62 five form two field families sharing stream prefixes (60 candidate rows recur across runs)—and per family: no non-degenerate rule matches more than one run in the NIST family or the X9.62 family (smallest-Hamming's two pooled matches split one and one), and the per-run matrix is archived with the reproduction package. No rule in the tested families reproduces the published selection in all ten runs: reverse last-viable matches all 10 only because it is true by construction, so it carries no explanatory information. Two limits are part of the result: a rule whose pick depends on candidates beyond the published one is untestable on the stored prefix by construction; and every battery statement is conditional on the single-run model. The primality arm's content is as stated in Method: every skipped-viable order's prime factor is certified prime, and a correct primality test never rejects a true prime, so a period test explains a skip only if it was buggy, nonstandard, or unfinished, or applied to a different input—for the $h > 1$ candidates, demanding N itself prime is the strict-tier filter, treated next. The strict tier, applied as a filter, explains exactly two runs—P-192 and P-521, whose skips all fail it; in the other eight, at least one skipped candidate passes even `V_strict`, and for P-256 the filter narrows the field to 876 and the published candidate, leaving 876 as the only unexplained skip. A transcription arm confirms byte-exact seed matches for the four NIST runs present in OpenSSL's seed list; P-256 and the X9.62 runs are absent from that list and unevaluated there.

The P-256 candidates at issue are 78 ($h = 4$), 876 ($h = 1$), and 1,842 ($h = 1$), all from the stream at $t = 871{,}696{,}354$; their seeds are printed in Result I and its footnotes.

The secondary question was searched within its bounds. B-163 is the one binary-curve seed the reconstruction did not place; the grid tested whether an `lrand48` stream at some 1997–98 timestamp produces it. The two priority windows (16 March–29 October 1997 and 2 February–16 June 1998; 31.2 million timestamps—$\pm$60 days around the seven committed session timestamps, enumerated in the reproduction package) cover the span of the other recovered runs; at each timestamp, candidate positions 1–200 were compared byte-for-byte against B-163's seed—a seed comparison, not a curve computation—and the position ceiling covers the recovered binary positions (32–155) with margin. No match: an exclusion within the searched region, conditional on the date-window assumption; the remaining hypothesis families (alternative generators, hashed passphrases, and the clock-divisor mappings—including the halved clock of Result III) are unrun; the grid, implementing the preregistration's primary hypothesis—the prime-field runs' parameterization, though Result III makes the halved clock the more plausible for a binary target—seeded the generator with the full timestamp. Two battery arms remain unevaluated: the wall-clock race model needs per-candidate evaluation costs the enumeration never recorded; and the crash-and-restart grid was deferred by the preregistration's cost scoping. What settles the selection question is therefore stated exactly: within every rule family the stored data can test, across all ten runs, the pre-committed outcome stands—the selection rule is not explained by the documented criteria tested so far.

TABLE III
THE MUNDANE-RULE BATTERY: RUNS (OF TEN) IN WHICH EACH PRE-COMMITTED RULE'S PICK EQUALS THE PUBLISHED CANDIDATE. A RULE EXPLAINS THE DATA ONLY AT 10 OF 10. THE REVERSE ROW IS A TAUTOLOGY CHECK, NOT A CANDIDATE EXPLANATION.

Rule (pick over each run's viable candidates)	Runs matched
Forward first-viable (smallest n)	0
Reverse last-viable (largest n)	10 (by construction)
Strided scan, stride 2, phases 0 / 1	0 / 0
Strided scan, stride 3, phases 0 / 1 / 2	1 / 0 / 0
Lexicographically smallest seed	1
Lexicographically largest seed	1
Smallest seed Hamming weight	2
Largest seed Hamming weight	0
Median viable candidate	0

SCOPE AND LIMITS

The accounting: replay, enumeration, and battery total $\approx$86 CPU-hours against the preregistration's 100-hour cap; the B-163 grid ran under its own 80-hour cap; control and revision-added computations were accounted outside both caps.

The P-256 claims in Result I rest on this audit's targeted verification, which the enumeration and battery figures agree with on P-256.

Three hedges bind every claim in this article. First, a refuted first-viable rule is not evidence of a backdoor: selection among viable candidates does not imply a weak curve was chosen, and no weakness in P-256 is claimed or implied. Second, a verified mundane process is not proof that nothing was selected: our audit cannot see unrecorded human choices, or a process that generated each seed in its own invocation, as Result III's binary sessions did. The skip inventories for the other runs are the enumeration's (Result IV); Anthropic's published counts differ from ours because the viability criteria differ; Anthropic does not publish theirs, so the totals cannot be reconciled criterion-by-criterion—though for P-256 the inventories agree on the load-bearing point, candidate 876. The run timestamp and candidate-indexing convention for all

runs follow Anthropic's reconstruction; the root representative is the one all ten published anchors match (Method); our verification is conditional on those inputs—we verify that (t, n) regenerates the published seed and coefficient, not the timestamp recovery itself, which no source we cite independently corroborates. The anchor is the seed, not the timestamp's provenance: a different stream start reproducing the same 160-bit seed at the published index is a coincidence of order 2^{-160}. Valsorda's reproduction agrees with the inventory but itself starts from Anthropic's recovered timestamps, so it corroborates everything downstream of them, not the recovery. Separately, the base point G and the order-selection step lie outside the verified pipeline, and the battery is silent about any selection that operated there. For candidate 876 the gap cannot bite: its order is prime, so every non-identity point generates the group, and no base-point choice changes the properties Table I checks. Third, inconclusive results are never framed as insinuation—where the evidence ends, the article says so, in the exact words committed before the data arrived.

Discussion: A Template for Verifiable Randomness

The P-256 refutation is a small fact: one viable candidate under the strict tier—876—was passed over, and the strictest criteria we apply do not explain why. Its significance lies less in the single skip than in what the audit demonstrates. Verifiable randomness is a claim about a *process*, and processes can be audited—but only if the audit is held to account: what was tested, what was not, and what each outcome meant before the data arrived.

Three practices generalize beyond this case. First, reproduce the derivation; do not trust the description: without byte-exact seed-to-coefficient matches across ten selections, the order computations would be arithmetic about an unverified pipeline. Second, pin the pipeline's edge cases explicitly: the $p \equiv 3 \pmod 4$ shortcut that broke on P-224 is the kind of silent assumption that voids a forensic claim, found only because verification was required against *every* published selection. Third, give null results a pre-committed home: "selection rule not explained by the tested criteria" is a publishable, checkable outcome, whereas an unplanned null invites the insinuation the hedges forbid.

New standards will again ask the world to trust generated parameters, and parameter generation should be documented to the level this audit required: the candidate space, the derivation code, the seeds, and the selection rule. Had the 1997 process recorded which rule chose among viable candidates, there would be no paper to write. The cost at generation time is small—a log file, a published script; the forensic cost of its absence is measured in decades.

Conclusion

The verified core—a refuted first-viable rule for P-256 (within the reconstructed stream and under the tiers of Scope and Limits), a byte-exact reconstruction pipeline covering ten published selections, a binary-curve control, a completed enumeration, and a rule battery over every family the stored data can test—is complete and archived. (Two battery arms remain unevaluated, as Result IV states.) For practitioners: no weakness in P-256 is shown, and no action is warranted. What follows is for the bodies and implementers who will generate the next shared parameters. The work needed is institutional rather than technical: make verifiable randomness auditable by construction, so the next audit confirms the record instead of reconstructing it. Standards bodies should require, as part of any parameter-generation submission, the six artifacts this audit had to reverse-engineer: (1) the candidate-generation program and generator implementation, with exact inputs and invocation boundaries recorded; (2) the viability predicate applied to each candidate; (3) the deterministic selection rule over the viable set; (4) the full ordered candidate list, not just the winner; (5) the evaluation log—per-candidate costs, timeouts, and restarts—whose absence here leaves the timing family of Result IV undecidable; (6) the base-point generation record. The template proposed here—exercised in its frozen definitions and pre-committed negatives—is the contribution we offer.

Data and Code Availability

The preregistration, analysis code, frozen data files, primality certificates and branch-control results of Method, and a one-command verifier for the headline numbers are archived in the reproduction package accompanying this article, at https://github.com/pchi1606/p256-seed-forensic-audit and on Zenodo under concept DOI 10.5281/zenodo.23478055; the preregistration is a separate deposit, DOI 10.5281/zenodo.23523983 [11]. The outputs are recomputable from the archived materials; the arms marked unevaluated are, by that marking, not claimed.

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